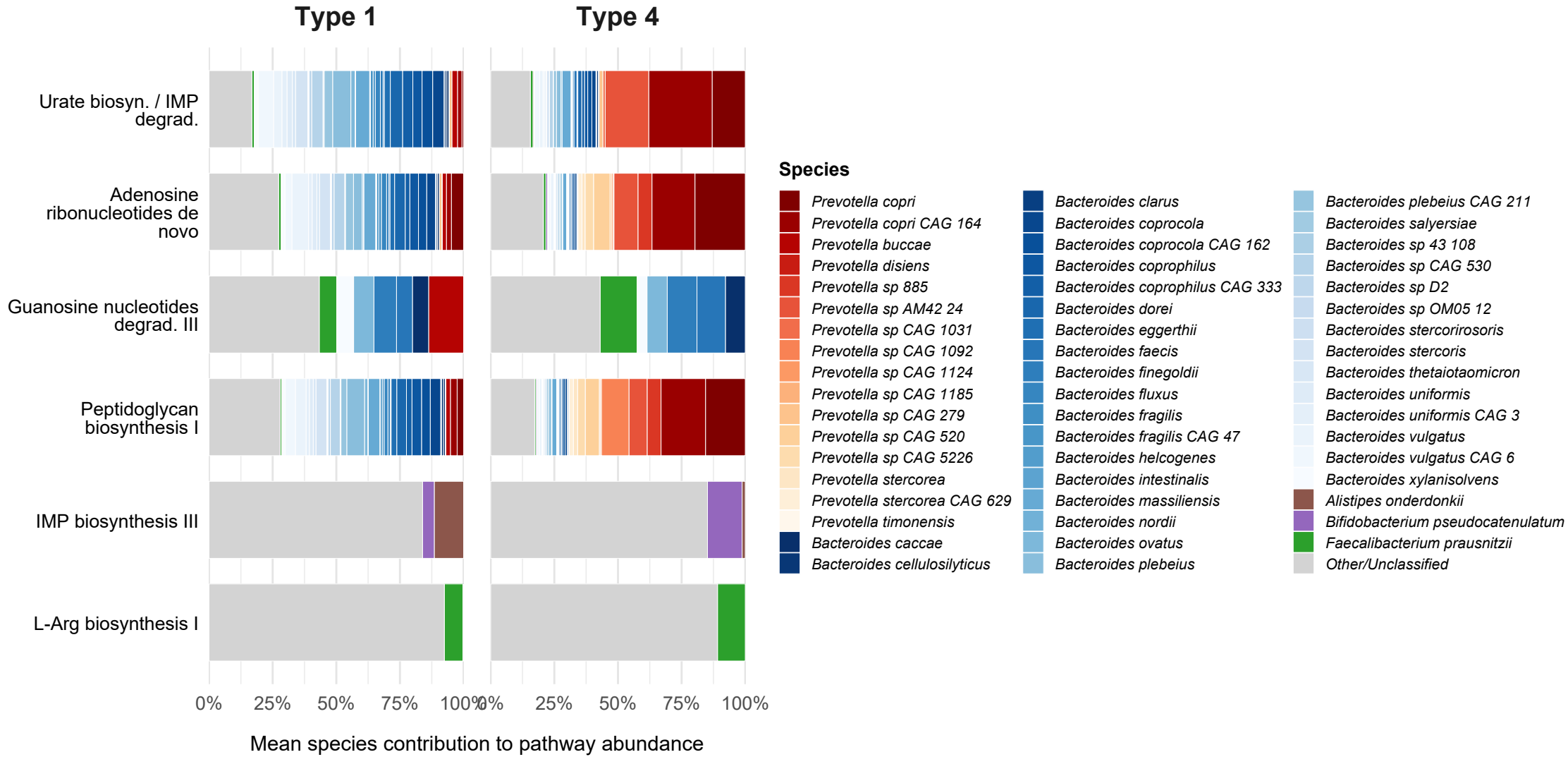
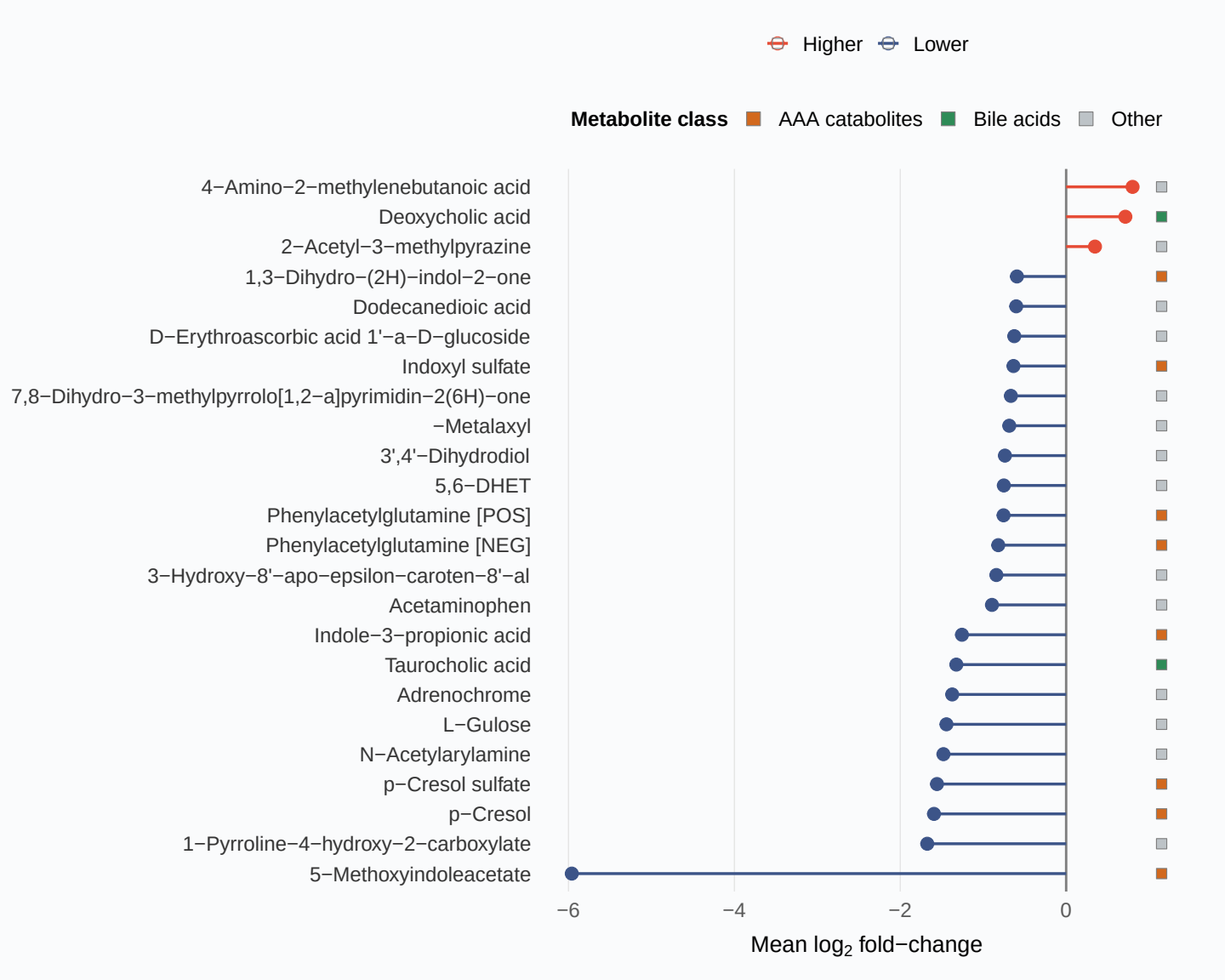
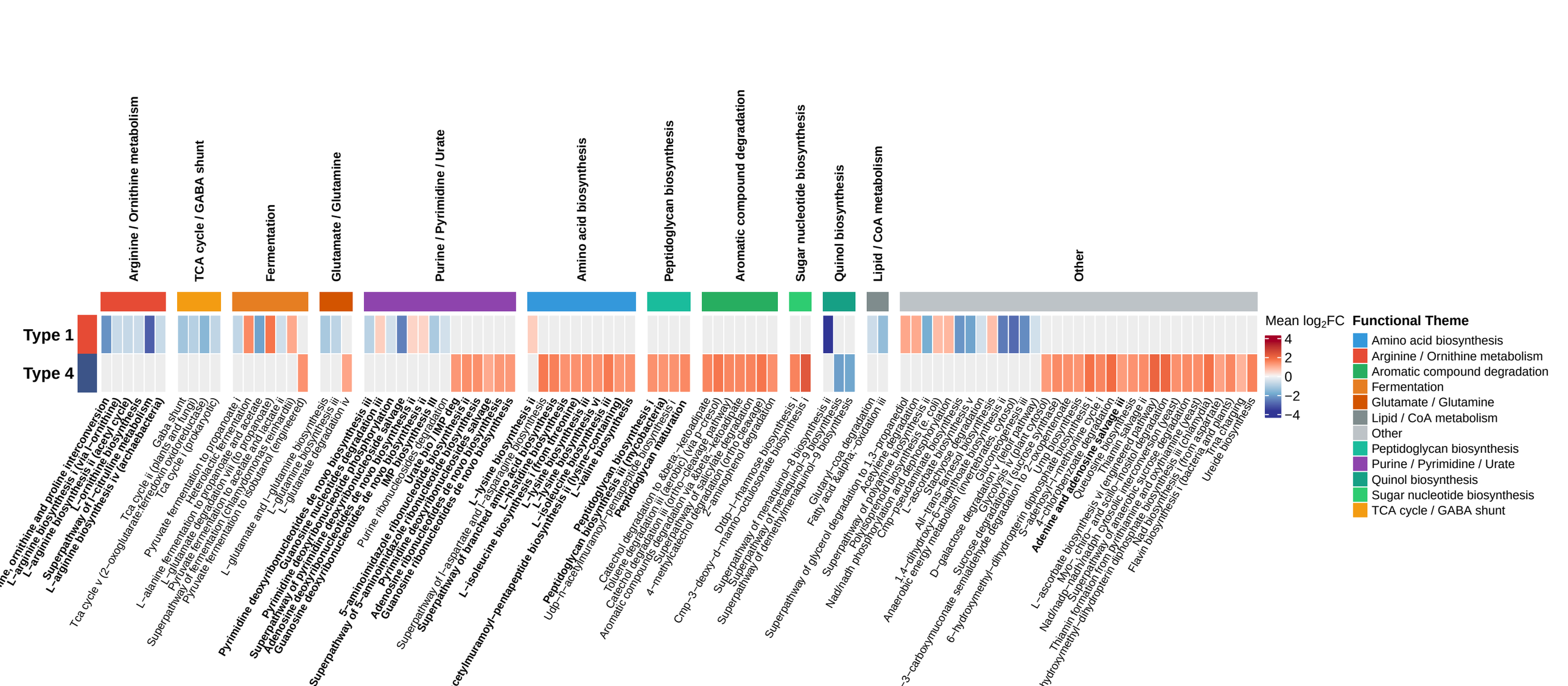

Prevotella copri
Roseburia sp CAG 471
Veillonella dispar
Collinsella aerofaciens
Adlercreutzia equolifaciens
Gordonibacter pamelaeae
Eggerthella lenta
Haemophilus parainfluenzae
Haemophilus sputorum
Desulfovibrio piger
Parasutterella excrementihominis
Fusobacterium varium
Fusobacterium mortiferum
Ruminococcus gnavus
Veillonella parvula
Erysipelatoclostridium ramosum
Clostridium bolteae
Clostridium symbiosum
Tyzzerella nexilis
Blautia hansenii
Veillonella infantium
Clostridium innocuum
Veillonella atypica
Blautia sp CAG 257
Streptococcus mitis
Hungatella hathewayi
Anaerotignum lactatifermentans
Flavonifractor plautii
Clostridium aldenense
Intestinibacter bartlettii
Eubacterium sulci
Clostridium clostridioforme
Clostridium lavalense
Holdemanella biformis
Clostridium sp CAG 242
Eubacterium sp CAG 180
Roseburia inulinivorans
Eubacterium hallii
Ruminococcus torques
Dorea longicatena
Dorea formicigenerans
Blautia obeum
Coprococcus catus
Eubacterium siraeum
Ruminococcus lactaris
Firmicutes bacterium CAG 95
Clostridium leptum
Gemmiger formicilis
Coprococcus comes
Firmicutes bacterium CAG 83
Firmicutes bacterium CAG 110
Ruminococcus bicirculans
Roseburia hominis
Oscillibacter sp CAG 241
Eubacterium rectale
Oscillibacter sp 57 20
Bacteroides fragilis
Bacteroides thetaiotaomicron
Parabacteroides distasonis
Bacteroides salyersiae
Paraprevotella xylaniphila
Bacteroides eggerthii
Parabacteroides merdae
Butyricimonas synergistica
Parabacteroides goldsteinii
Bacteroides caccae
Bacteroides intestinalis
Alistipes shahii
Barnesiella intestinihominis
Butyricimonas virosa
Alistipes finegoldii
Bacteroides plebeius
Odoribacter splanchnicus
Alistipes putredinis
Core Status
Shared
Type 1 Exclusive
Type 4 Exclusive
Phylum
Actinobacteria
Bacteroidetes
Firmicutes
Fusobacteria
Proteobacteria


Type 1
Type 4
Urate biosyn. / IMP degrad.
Adenosine ribonucleotides de novo
Guanosine nucleotides degrad. III
Peptidoglycan biosynthesis I
IMP biosynthesis III
L-Arg biosynthesis I
0%
25%
50%
75%
100%
Mean species contribution to pathway abundance
Species
Prevotella copri
Prevotella copri CAG 164
Prevotella buccae
Prevotella disiens
Prevotella sp 885
Prevotella sp AM42 24
Prevotella sp CAG 1031
Prevotella sp CAG 1092
Prevotella sp CAG 1124
Prevotella sp CAG 1185
Prevotella sp CAG 279
Prevotella sp CAG 520
Prevotella sp CAG 5226
Prevotella stercorea
Prevotella stercorea CAG 629
Prevotella timonensis
Bacteroides caccae
Bacteroides cellulosilyticus
Bacteroides clarus
Bacteroides coprocola
Bacteroides coprocola CAG 162
Bacteroides coprophilus
Bacteroides coprophilus CAG 333
Bacteroides dorei
Bacteroides eggerthii
Bacteroides faecis
Bacteroides finegoldii
Bacteroides fluxus
Bacteroides fragilis
Bacteroides fragilis CAG 47
Bacteroides helcogenes
Bacteroides intestinalis
Bacteroides massiliensis
Bacteroides nordii
Bacteroides ovatus
Bacteroides plebeius
Bacteroides plebeius CAG 211
Bacteroides salyersiae
Bacteroides sp 43 108
Bacteroides sp CAG 530
Bacteroides sp D2
Bacteroides sp OM05 12
Bacteroides stercorirosoris
Bacteroides stercoris
Bacteroides thetaiotaomicron
Bacteroides uniformis
Bacteroides uniformis CAG 3
Bacteroides vulgatus
Bacteroides vulgatus CAG 6
Bacteroides xylanisolvens
Alistipes onderdonkii
Bifidobacterium pseudocatenulatum
Faecalibacterium prausnitzii
Other/Unclassified


Higher
Lower
Metabolite class
AAA catabolites
Bile acids
Other
4−Amino−2−methylenebutanoic acid
Deoxycholic acid
2−Acetyl−3−methylpyrazine
1,3−Dihydro−(2H)−indol−2−one
Dodecanedioic acid
D−Erythroascorbic acid 1'−a−D−glucoside
Indoxyl sulfate
7,8−Dihydro−3−methylpyrrolo[1,2−a]pyrimidin−2(6H)−one
−Metalaxyl
3',4'−Dihydrodiol
5,6−DHET
Phenylacetylglutamine [POS]
Phenylacetylglutamine [NEG]
3−Hydroxy−8'−apo−epsilon−caroten−8'−al
Acetaminophen
Indole−3−propionic acid
Taurocholic acid
Adrenochrome
L−Gulose
N−Acetylarylamine
p−Cresol sulfate
p−Cresol
1−Pyrroline−4−hydroxy−2−carboxylate
5−Methoxyindoleacetate
−6
−4
−2
0
Mean log2 fold−change


Arginine / Ornithine metabolism
TCA cycle / GABA shunt
Fermentation
Glutamate / Glutamine
Purine / Pyrimidine / Urate
Amino acid biosynthesis
Peptidoglycan biosynthesis
Aromatic compound degradation
Sugar nucleotide biosynthesis
Quinol biosynthesis
Lipid / CoA metabolism
Other
Type 1
Type 4
Mean log2FC
4
2
0
−2
−4
Functional Theme
Amino acid biosynthesis
Arginine / Ornithine metabolism
Aromatic compound degradation
Fermentation
Glutamate / Glutamine
Lipid / CoA metabolism
Other
Peptidoglycan biosynthesis
Purine / Pyrimidine / Urate
Quinol biosynthesis
Sugar nucleotide biosynthesis
TCA cycle / GABA shunt
L−arginine biosynthesis i (via l−ornithine)
L−arginine biosynthesis ii (acetyl cycle)
Superpathway of l−citrulline metabolism
L−arginine biosynthesis iv (archaebacteria)
Tca cycle v (2−oxoglutarate:ferredoxin oxidoreductase)
Tca cycle ii (plants and fungi)
Gaba shunt
Tca cycle i (prokaryotic)
Pyruvate fermentation to propanoate i
L−alanine fermentation to propanoate and acetate
L−glutamate degradation to propanoate and acetate ii
Superpathway of fermentation (to propanoate and lactate ii)
Pyruvate fermentation to acetate and lactate ii
Pyruvate fermentation (chlamydomonas reinhardtii)
Pyruvate fermentation to isobutanol (engineered)
L−glutamate and l−glutamine biosynthesis iii
L−glutamate biosynthesis iii
L−glutamate degradation iv
Pyrimidine deoxyribonucleotides de novo biosynthesis iii
Pyrimidine deoxyribonucleotides degradation iii
Guanosine deoxyribonucleotides phosphorylation
Superpathway of pyrimidine deoxyribonucleotide salvage
Adenosine deoxyribonucleotides de novo biosynthesis ii
Guanosine deoxyribonucleotides de novo biosynthesis ii
Superpathway of 5−aminoimidazole ribonucleotide biosynthesis
5−aminoimidazole ribonucleotide biosynthesis i
Pyrimidine deoxyribonucleotide biosynthesis
Adenosine ribonucleotides de novo biosynthesis
Guanosine ribonucleotides de novo biosynthesis
Superpathway of branched amino acid biosynthesis
Purine ribonucleosides degradation
IMP biosynthesis ii
IMP biosynthesis iii
Urate biosynthesis/inosine 5'−phosphate degradation
IMP deg
Superpathway of l−aspartate and l−asparagine biosynthesis
L−lysine biosynthesis ii
L−isoleucine biosynthesis i (from threonine)
L−histidine biosynthesis
L−lysine biosynthesis i
L−isoleucine biosynthesis ii
L−lysine biosynthesis vi
L−isoleucine biosynthesis iii (lysine−containing)
L−valine biosynthesis
Peptidoglycan biosynthesis i
Peptidoglycan biosynthesis iii (mycobacteria)
Peptidoglycan maturation
Udp−n−acetylmuramoyl−pentapeptide biosynthesis i
Catechol degradation to 2−hydroxypentadienoate i
Toluene degradation iii (aerobic) (via p−cresol)
Catechol degradation iii (ortho−cleavage pathway)
Aromatic compounds degradation via beta−ketoadipate
Superpathway of salicylate degradation
4−methylcatechol degradation (ortho−cleavage)
Cmp−3−deoxy−d−manno−octulosonate biosynthesis i
Superpathway of menaquinol−8 biosynthesis ii
Superpathway of menaquinol−9 biosynthesis
Superpathway of demethylmenaquinol−9 biosynthesis i
Glutaryl−coa degradation
Fatty acid α−oxidation iii
Superpathway of glycerol degradation to 1,3−propanediol
Acetylene degradation
Superpathway of polyamine biosynthesis ii
Polyisoprenoid biosynthesis (e. coli)
Nad/nadh phosphorylation and dephosphorylation
Cmp−pseudaminate biosynthesis
L−ascorbate biosynthesis v
Stachyose degradation
All−trans−farnesol biosynthesis
1,4−dihydroxy−6−naphthoate biosynthesis ii
Anaerobic energy metabolism (invertebrates, cytosol)
D−galactose degradation v (leloir pathway)
Sucrose degradation iv (sucrose phosphorylase)
Glycolysis v (plant cytosol)
Gluconeogenesis ii (plant cytosol)
Superpathway of l−methionine biosynthesis i
5−aminoimidazole ribonucleotide biosynthesis i
4−chlorobenzoate degradation
Ump biosynthesis
Thiamin salvage ii
Queuosine biosynthesis
L−ascorbate biosynthesis vi (engineered pathway)
Myo−, chiro− and scyllo−inositol degradation (yeast)
Nad/nadp−nadh/nadph cytosolic interconversion (yeast)
Thiamin formation from pyrithiamine and oxythiamine (yeast)
Superpathway of anaerobic sucrose degradation
Hydroxymethyl−dihydropterin diphosphate biosynthesis i (from aspartate)
Nad biosynthesis i (chlamydia)
Flavin biosynthesis i (bacteria and plants)
Trna charging
Ureide biosynthesis
Cmp−3−deoxy−d−manno−oct−2−ulosonate biosynthesis
6−hydroxymethyl−dihydropterin diphosphate biosynthesis iii (chlamydia)
Adenine and adenosine salvage iii